

LAB ACTIVITY 3 – PART 1: THE DATA FOUNDATION

Topic: 3.1.1 JavaScript Variables, Data Types, and Operators

Duration: 45–60 Minutes

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Date	26/2/2026

1. Learning Outcomes

By the end of this session, you should be able to:

1. **Declare** variables using `let` and `const` appropriately.
2. **Identify** different JavaScript data types (String, Number, Boolean, Array).
3. **Perform** calculations using Arithmetic and Assignment operators.
4. **Display** data using `console.log()` and `alert()`.

2. Part A: Theory (The Building Blocks)

Concept	Description	Example
Variables	Containers for storing data values.	<code>let price = 10;</code>
Strings	Textual data wrapped in quotes.	<code>"Hello World"</code>
Numbers	Numerical data (integers or decimals).	<code>25</code> or <code>3.14</code>
Booleans	Logical values: True or False.	<code>isLoggedIn = true;</code>
Operators	Symbols used to perform operations.	<code>+</code> , <code>-</code> , <code>*</code> , <code>/</code> , <code>%</code>

3. Part B: Practical Step-by-Step

Step 1: Variable Declaration & Setup

Goal: Set up the file structure and define basic variables.

1. Create a folder named **RegNo_Lab3_Part1**.

2. Create an index.html file.
3. Inside the <body>, insert the Header and the first Lab Card:

HTML

```
<header>
  <h1>Lab 3.1.1: JavaScript Fundamentals</h1>
  <p>Variables, Data Types, and Operators</p>
</header>

<section class="lab-card">
  <h2>Steps 1 - 3: Logic & Console</h2>
  <div class="instruction">
    Open the <strong>Browser Console</strong> (F12 > Console) to see the output for variables
    and <code>typeof</code> checks.
  </div>
  <button onclick="runConsoleTasks()">Run Console Tasks</button>
  <p class="note">*This logs Course Name, Student Count, and Data Types.</p>
</section>
```

4. Inside the <script> tag, define the first function covering **Steps 1, 2, and 3**:

JavaScript

```
/* =====
   STEPS 1, 2 & 3: Variables & Operators
   ===== */
function runConsoleTasks() {
  // Variable Declaration
  const courseName = "Web Design Technology";
  let studentCount = 30;
  let isLabFinished = false;

  console.log("--- Step 1: Variables ---");
  console.log("Course:", courseName);
  console.log("Initial Students:", studentCount);

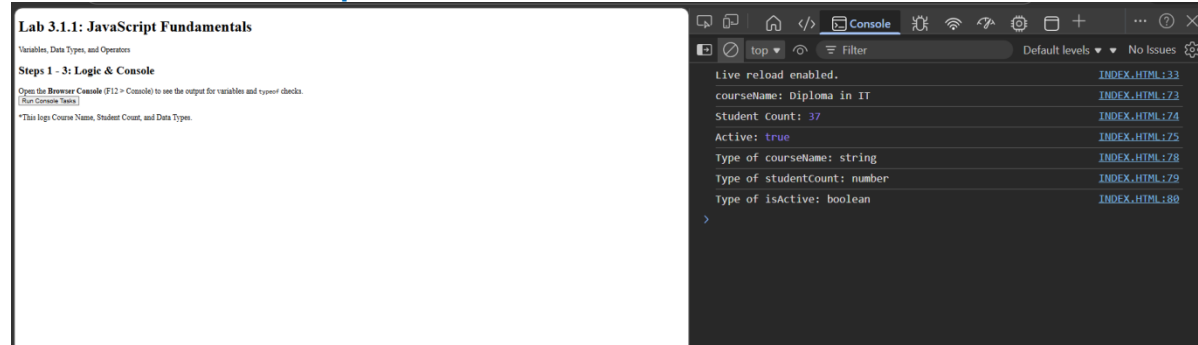
  // Arithmetic
  let x = 10;
  let y = 3;
  console.log("--- Step 2: Arithmetic ---");
  console.log("Addition (10+3):", x + y);
  console.log("Remainder (10%3):", x % y);

  // Type Identification
  console.log("--- Step 3: Data Types ---");
  console.log("Type of courseName:", typeof courseName);
  console.log("Type of studentCount:", typeof studentCount);
  console.log("Type of isLabFinished:", typeof isLabFinished);

  alert("Check your browser console (F12) to see the results!");
}
```

5. Capture a screenshot of the output from the previous steps. **Observe** the results and **explain** your understanding of the process.

Screenshot of the output



Explanation

The values printed course name, student count, isActive

Step 4: Simple UI Interaction (Prompt & Alert)

Goal: Interact with the user using built-in browser dialog boxes.

1. Add this new section to your HTML <body>:

HTML

```
<section class="lab-card">
  <h2>Step 4: Simple UI Output</h2>
  <div class="instruction">
    Test the <code>prompt()</code>, <code>alert()</code>, and <code>confirm()</code> functions.
  </div>
  <button onclick="runInteraction()">Start Interaction</button>
</section>
```

2. Add the function to your <script> tag (below runConsoleTasks):

JavaScript

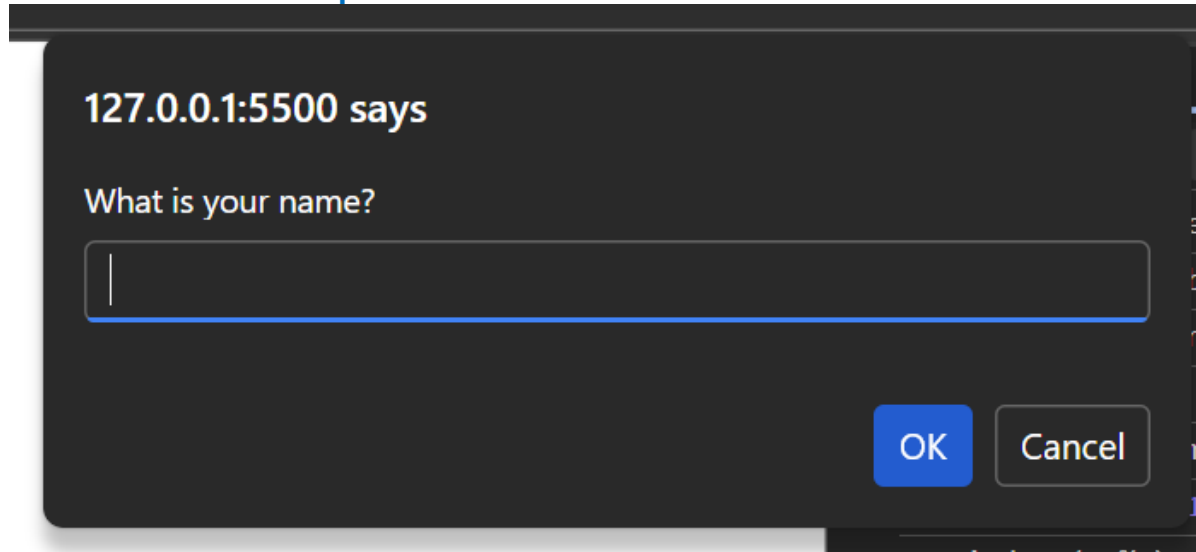
```

/* =====
STEP 4: Interaction
===== */
function runInteraction() {
    let username = prompt("What is your name?");
    if (username) {
        alert("Welcome to JS Lab, " + username + "!");

        let proceed = confirm("Do you want to start the lab?");
        console.log("User wants to proceed:", proceed);
    }
}

```

Screenshot of the output



Explanation

prompt() collects text input

4. Final Practical Task: The POS Script

Scenario: Construct a "Point-of-Sale" script to calculate a total bill including a 6% service tax.

1. Add the HTML UI (The Trigger):

HTML

```

<section class="lab-card" style="border: 2px solid #27ae60; margin-top: 20px; padding: 15px; border-radius: 8px;">
  <h2 style="color: #27ae60;">Final Task: POS Script</h2>
  <div class="instruction">
    <strong>Scenario:</strong> Calculate a total bill with 6% Service Tax.
  </div>
  <p>Price per Item: <strong>RM 15.00</strong></p>

```

```

<p>Tax Rate: <strong>6%</strong></p>

<button onclick="runPOS()" style="background-color: #27ae60; color: white; padding: 10px 20px;
border: none; border-radius: 5px; cursor: pointer;">
  Checkout Now
</button>
</section>

```

2. Add the JavaScript Logic:

JavaScript

```

/* =====
FINAL TASK: Point-of-Sale (POS)
===== */
function runPOS() {
  // Constants
  const pricePerItem = 15;
  const taxRate = 0.06;

  // Input
  let quantity = prompt("How many items do you want to buy?");

  // Validation & Calculation
  if (quantity !== null && quantity !== "") {
    let qtyNumber = parseInt(quantity);

    if (!isNaN(qtyNumber) && qtyNumber > 0) {
      let subtotal = pricePerItem * qtyNumber;
      let totalTax = subtotal * taxRate;
      let finalBill = subtotal + totalTax;

      // Output with 2 decimal places
      alert("Order Summary:\n" +
        "-----\n" +
        "Quantity: " + qtyNumber + "\n" +
        "Subtotal: RM " + subtotal.toFixed(2) + "\n" +
        "Service Tax (6%): RM " + totalTax.toFixed(2) + "\n\n" +
        "Your total bill is: RM " + finalBill.toFixed(2));
    } else {
      alert("Invalid input! Please enter a valid number.");
    }
  }
}

```

What happens when you run this?

1. **Prompt:** Input box pops up. Type 3.
2. **Calculation:** * Subtotal:

$\$3 \times 15 = 45\$$

- o Tax:

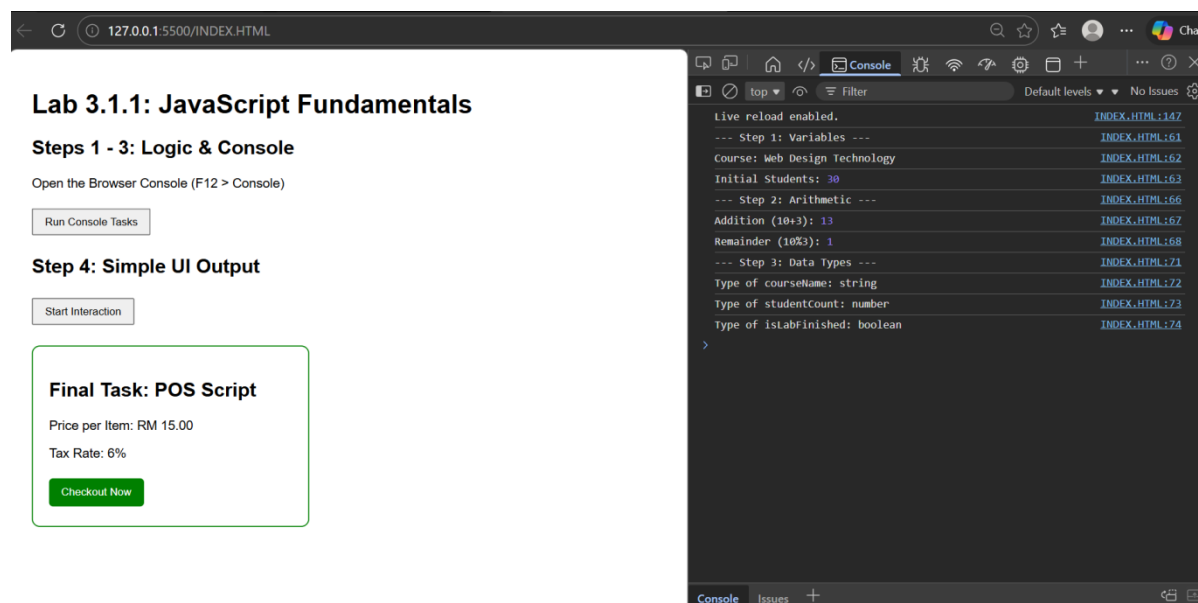
$\$45 \times 0.06 = 2.70\$$

- o Total:

$\$45 + 2.70 = 47.70\$$

3. **Output:** An alert box appears: **"Your total bill including tax (6%) is: RM 47.70"**.

Screenshot of the output



Explanation

Example with quantity = 3:

Subtotal:

$$3 \times 15 = 45 \quad 3 \times 15 = 45$$

Tax:

$$45 \times 0.06 = 2.7045 \quad 45 \times 0.06 = 2.7045$$

Total:

$$45 + 2.70 = 47.7045 \quad 45 + 2.70 = 47.7045$$

5. Quick Check Questions

Part A

Question 1

Which of the following correctly describes the difference between the = and + operators in JavaScript?

- A. = is used for addition, while + is used to assign values to variables.
- B. = is the Assignment Operator (stores a value), while + is the Arithmetic or Concatenation Operator.
- C. Both are Arithmetic Operators used for mathematical calculations only.
- D. = compares two values, while + creates a new variable.

Question 2

What is the result of the expression "5" * 2 in JavaScript, and why?

- A. 55, because JavaScript repeats the string.
- B. Error, because you cannot multiply a string with a number.
- C. 10, because JavaScript performs "Type Coercion" to convert the string to a number.
- D. NaN (Not a Number), because the operation is mathematically impossible.

Question 3

In a Point-of-Sale (POS) system, why is it recommended to use const instead of let for a variable like taxRate = 0.06?

- A. const allows the tax rate to be changed easily later in the script.
- B. const makes the variable run faster in the browser memory.
- C. const prevents accidental reassignment, ensuring the fixed tax value remains consistent.
- D. const is the only keyword that supports decimal (float) numbers.

Part B

1. Which keyword is used to declare a variable whose value is intended to remain constant throughout the program?

- A. let
- B. var
- C. **const**
- D. static

2. What is the output of the following code? `console.log(typeof "2026");`

- A. number
- B. **string**
- C. boolean
- D. undefined

3. Which operator is used to find the remainder of a division between two numbers?

- A. /
- B. *
- C. &
- D. **%**

4. How do you display a message box to the user that requires them to click "OK" or "Cancel"?

- A. alert()
- B. prompt()
- C. **confirm()**
- D. console.log()

5. Which of the following is a valid Boolean value in JavaScript?

- A. "true"
- B. Yes
- C. **false**
- D. 1

6. What will be the result of the following operation: let x = "5" + 2;?

A. 7

B. 52

C. NaN

D. Error

7. Which method is used to convert a string into a whole number (integer)?

A. parseFloat()

B. toString()

C. parseInt()

D. toNumber()

8. In JavaScript, which operator is used for "Strict Equality" (checking both value and data type)?

A. =

B. ==

C. ===

D. !==

9. What is the purpose of the prompt() function?

A. To display a warning message.

B. To print data to the browser console.

C. To allow the user to input text data.

D. To permanently delete a variable.

10. If let a = 10; and let b = 3;, what is the result of a ** b (Exponentiation)?

A. 13

B. 30

C. 100

D. 1000

💡 JavaScript Quick Tips for Students

Step 1: Variables (*const* vs *let*)

- **const (Constant):** Gunakan untuk nilai yang tidak akan berubah. Contohnya: Nama kursus atau Kadar Cukai. Jika anda cuba ubah nilainya, JavaScript akan bagi error.
- **let:** Gunakan untuk nilai yang boleh berubah-ubah. Contohnya: studentCount (bilangan pelajar boleh bertambah atau berkurang).
- **Boolean:** Hanya ada dua nilai: true atau false. Sangat berguna untuk buat keputusan (Logic).

Step 2: Arithmetic Operators

- **+, -, *, /:** Berfungsi seperti kalkulator biasa.
- **% (Modulus):** Ini bukan peratus! Ia mencari **baki** pembahagian.
 - Contoh: $10 \% 3 = 1$ (kerana $3 \times 3 = 9$, bakinya adalah 1).
 - Kegunaan: Selalu digunakan untuk semak nombor genap/ganjil.

Step 3: The *typeof* Operator

- JavaScript adalah bahasa *dynamically typed*. Maksudnya, kita tak perlu beritahu komputer jika pemboleh ubah itu teks atau nombor, JavaScript akan "teka" sendiri.
- Gunakan *typeof* untuk **debug** kod anda jika pengiraan matematik tidak menjadi (mungkin anda tertambah Teks dengan Nombor!).

Step 4: Interaction (*prompt*, *alert*, *confirm*)

- **prompt():** Cara paling mudah untuk dapatkan input dari pengguna. **Penting:** Input dari prompt sentiasa dalam bentuk **String (Teks)**.
- **alert():** Untuk beri informasi sehalu kepada pengguna.
- **confirm():** Menghasilkan nilai true (jika tekan OK) atau false (jika tekan Cancel).

Final Step: POS Logic (*The "Pro" Tips*)

- **parseInt():** Oleh kerana *prompt()* memberikan teks, kita perlu "paksa" ia menjadi nombor bulat menggunakan *parseInt()* supaya kita boleh buat pengiraan matematik.
- **isNaN() (Is Not a Number):** Fungsi keselamatan. Ia menyemak sama ada input pengguna itu nombor sah atau bukan (contoh: jika user taip "ABC", *isNaN* akan jadi true).
- **.toFixed(2):** Sangat penting untuk aplikasi kewangan. Ia memastikan nombor kita sentiasa ada 2 tempat perpuluhan (RM 10.00, bukan RM 10).



Lab 3.1.1: JavaScript Fundamentals Quiz

1. Which keyword should you use to declare a variable that will NOT change throughout the script (e.g., a Tax Rate)?

- A) let
- B) var
- C) **const**
- D) fixed

2. What is the result of the expression `10 % 4` in JavaScript?

- A) 2.5
- B) **2**
- C) 4
- D) 0

3. If you use `let input = prompt("Enter a number")` and the user types 5, what is the data type of input?

- A) Number
- B) Boolean
- C) Null
- D) **String**

4. What is the purpose of the `parseInt()` function in the POS script?

- A) To round a number to the nearest decimal.
- B) **To convert a string (text) into an integer (number).**
- C) To calculate the 6% service tax.
- D) To display a message on the screen.

5. Which method is used to format a number to exactly two decimal places (e.g., converting 15 to 15.00)?

A) toTwo()

B) math.round(2)

C) toFixed(2)

D) formatCurrency()

✓ Answer Key:

1. **C** (const is for constants)
2. **B** (10 divided by 4 is 2 with a **remainder of 2**)
3. **D** (All prompt inputs are returned as **strings** by default)
4. **B** (It ensures the "text" from the prompt can be used for math)
5. **C** (toFixed(2) handles the decimal formatting)